ABYLAIKHAN MUKHAMEJANOV

MECHANICAL ENGINEER

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Mechanical engineer with 3 years of hands-on experience designing, machining, and testing electromechanical hardware in Harvard and Lehigh labs, industry, and Formula SAE. Interested in hardware product development.

AREAS OF EXPERTISE

MECHANICAL DESIGN · PRODUCT DEVELOPMENT · DFM & DFA · GD&T · FEA · CNC MACHINING · 3D PRINTING

EDUCATION

Lehigh University · *B.S. Mechanical Engineering; GPA 3.64/4.00* — Bethlehem, PA · Aug. 2023 – May 2027

RELEVANT EXPERIENCE

Lauder Laboratory, Harvard University — May 2026 – Aug. 2026
Research Intern – Office of Naval Research MURI Program — *Cambridge, MA*

- Studied in-line swimming with a leading and a trailing robotic fish, showing how the trailing fish gains thrust from the leader's wake without actively swimming, with applications to ships and underwater drones.
- Operated an air-bearing robotic swimmer (LabVIEW, PID) and developed a test matrix to identify thrust-generating parameters.
- Used particle image velocimetry (PIV) to map water flow, visualize wake vortices, and study how the trailing fish gains thrust.
- Analyzed swimming kinematics in R and MATLAB, and documented test methods and findings in a technical report.

Unsteady Flow Interactions Laboratory, Lehigh University — Nov. 2023 – Present
Undergraduate Researcher – Mechanical Design & Test — *Bethlehem, PA*

- Designed and built a robotic swimmer system for a water channel to study energy and thrust generation from ocean waves under an Office of Naval Research program, with applications to ship propulsion and ocean energy generation.
- Built a cyber-physical platform that physically replicates programmed mass, stiffness, and damping, letting researchers change foil parameters in seconds instead of swapping springs and foils or reassembling hardware, saving setup time and cost.
- Designed 45+ custom parts in SolidWorks, applying GD&T and DFM/DFA for CNC machining, waterjet cutting, and 3D printing.
- Traced motion error to backlash through tolerance analysis and redesigned the assembly, reducing tracking error 5×.
- Integrated servos, encoders, and a six-axis force/torque sensor using NI DAQ and a MATLAB/Simulink control loop.
- Added emergency stops and motion and velocity limits, and wrote operating guidance enabling safe use by lab members.
- **Best Undergraduate Research Poster Award**, ASME IMECE 2025; presented at APS Fluid Dynamics and ONR MURI reviews.

Formula SAE Racing Team, Lehigh University — Dec. 2025 – Present
Design and Manufacturing Engineer — *Bethlehem, PA*

- Redesigned steel suspension control arms by switching to carbon-fiber tubes bonded to aluminum joints, reducing unsprung mass.
- Sized joints in ANSYS FEA for braking and cornering loads with a minimum safety factor of 2.0, and developed the team FMEA.
- Designed joints for manufacture and assembly with a controlled epoxy bond gap and O-ring seal.
- Machining 6061-T6 aluminum joints in-house using CNC milling and turning, holding drawing tolerances on bond and seal surfaces.
- Wrote a 45-specimen bonded-joint test plan (bond type, length, surface treatment) to validate the design before it goes on the car.

Aerotargets International LLC — May 2024 – Aug. 2024
Mechanical Engineering Intern — *Kingston, PA*

- Ran wind-tunnel tests on a 33% scale UAV model up to 84 mph, Phase 3 of a three-phase test program.
- Designed a SolidWorks fixture mounting the model on a six-axis force/torque sensor, with a four-bar linkage to adjust test angles without re-rigging.
- Analyzed force data using NI DAQ and MATLAB and recommended fuselage panel design changes that lowered drag and raised lift-to-drag ratio about 10%.

TEACHING & COMMUNITY IMPACT

Head Teaching Assistant, ENGR 010 – Lehigh University — Dec. 2023 – Present

- Lead a 30-student section in weekly Python and robotics labs; authored 8 hands-on robotics lab modules.
- Coordinate a team of 3–4 TAs for the section and mentor 1–2 new TAs each semester.

Makerspace Coordinator – Lehigh University — Aug. 2024 – Jan. 2025

- Supported fabrication of 30+ projects using 3D printers, CNC routers, waterjet cutters, and laser cutters.

TECHNICAL SKILLS

SolidWorks · Fusion 360 · ANSYS FEA · Mastercam · Engineering Drawings (ASME Y14.5) · DFMA · CNC Milling & Turning
NI DAQ · LabVIEW · Arduino · Raspberry Pi · MATLAB · Simulink · PID Control · Root Cause Analysis · Python · C++ · R · Excel

Engineering Portfolio: abylaikhanm.github.io